

# Generic Steklov eigenfunctions and local variation of edge weights

Kun Yang\*

## Abstract

Let  $G$  be a finite connected graph with a designated boundary  $B$ . We prove that, for generic positive edge weights, every nonconstant Steklov eigenfunction is injective on the boundary, and its harmonic extension is injective on the union of simple boundary-to-boundary paths. More generally, every preassigned finite family of homogeneous polynomial relations is avoided on this core unless a relation follows from the zero boundary sum. The proof is based on an exact local rank formula. At a simple positive eigenvalue, the differential from a prescribed family of edge weights to the eigenvalue and the normalized extension has rank  $|V(G)| - c$ , where  $c$  is the number of components formed by the variable non-flat edges. Consequently, the weights on any non-flat spanning tree of the core are local analytic coordinates for one normalized eigenpair, with a rational inverse. Joint polynomial avoidance and absolute continuity under random weights follow as corollaries. We conclude with a positive-flow criterion describing the full-extension data realizable by positive weights and the freedom in those weights.

**Keywords:** Steklov eigenfunction, weighted graph, generic injectivity, harmonic extension, local variation.

## 1 Introduction and main results

Generic properties of eigenfunctions have been studied since the work of Uhlenbeck [11]. For the Steklov problem on a smooth compact manifold, Wang [12] proved generic simplicity and generic regularity properties of the boundary eigenfunctions. The discrete Steklov problem was studied by Hua, Huang and Wang [6]. More recently, Chen, Hua and Shi [3] proved that, for generic positive edge weights on a fixed finite graph, the Steklov spectrum is simple and every eigenfunction is nonzero at every boundary vertex. Their proof combines algebraic exceptional sets with a construction by pendant attachments.

We consider two questions beyond boundary nonvanishing. Which relations between the values of an eigenfunction are forced by the graph? How many of these values can be varied independently by changing only a prescribed collection of edge weights? The first question leads to a generic polynomial-avoidance theorem on the boundary core. The second leads to an exact differential-rank formula, which supplies the local freedom needed in the proof of the first. The formula separates the spectral issue, handled by simplicity, from the remaining obstruction, which is exactly the connectivity of the variable non-flat edges.

Throughout,  $G = (V, E)$  is a finite connected simple graph,  $B \subseteq V$  is nonempty, and  $w \in \mathbb{R}_{>0}^E$  is a vector of positive edge weights. The boundary need not be independent or consist of leaves.

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\*School of Mathematical Sciences, Nankai University, Tianjin 300071, China.  
Email: 2313878@mail.nankai.edu.cn.

Write  $I = V \setminus B$ . The weighted Laplacian is

$$(L(w)u)(x) = \sum_{y \sim x} w(xy)(u(x) - u(y)).$$

A Steklov eigenpair satisfies

$$L(w)u = \lambda Mu, \quad M = \text{diag}(\mathbf{1}_B), \quad u|_B \neq 0. \quad (1.1)$$

Thus  $u$  is harmonic on  $I$  and its outward normal derivative on  $B$  equals  $\lambda u|_B$ . We call  $f = u|_B$  the boundary eigenfunction and  $u$  its extension. The Steklov eigenvalues are  $0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_{|B|-1}$ ; the zero eigenspace consists of constants. Every extension for a positive eigenvalue satisfies  $\sum_{b \in B} u(b) = 0$ . These facts are recalled in Section 2. When  $|B| = 1$ , there is no positive eigenvalue, so the assertions concerning nonconstant eigenfunctions are empty. Unless otherwise stated,  $|B| \geq 2$ .

The  $B$ -core  $G^B$  is the union of all simple paths whose endpoints belong to  $B$ . Paths of length zero are allowed. Write  $V_0 = V(G^B)$  and  $N_0 = |V_0|$ . Every component of  $G - V_0$  has a unique attachment vertex in  $V_0$ , and every harmonic extension is constant on that component. Moreover, the Steklov matrix depends only on the core weights. We prove these statements in Lemma 2.1. The core is therefore the natural domain for injectivity.

Here a property holds for *generic positive weights* if its exceptional set is contained in the zero set of one nonzero polynomial in  $\mathbb{R}[w_e : e \in E]$ . The property then holds on a dense open set of full Lebesgue measure. Countable unions of such exceptional sets will be stated separately.

## 1.1 Polynomial relations on the core

Write

$$\mathcal{H}_B = \left\{ v \in \mathbb{R}^{V_0} : \sum_{b \in B} v(b) = 0 \right\}.$$

Relations between unnormalized eigenfunctions must respect multiplication by a nonzero scalar. We therefore first consider homogeneous polynomials.

**Theorem 1.1.** *Let  $\mathcal{F}$  be a finite family of homogeneous polynomials in  $\mathbb{R}[x_v : v \in V_0]$ , none vanishing identically on  $\mathcal{H}_B$ . There is a nonzero polynomial  $R_{\mathcal{F}}$  in the edge weights such that  $R_{\mathcal{F}}(w) \neq 0$  implies a simple Steklov spectrum and*

$$Q(u|_{V_0}) \neq 0$$

*for every  $Q \in \mathcal{F}$  and every extension  $u$  belonging to a positive Steklov eigenvalue.*

*For an at most countable family with the same hypothesis, the exceptional weights are contained in a countable union of zero sets of nonzero polynomials. A homogeneous polynomial vanishes on every such extension for every positive weight if and only if it is divisible by  $\sum_{b \in B} x_b$ .*

The family in this theorem is prescribed before the weights are chosen. In particular, the theorem does not exclude all real-coefficient polynomials simultaneously. Its countable assertion does allow all nontrivial homogeneous relations with rational coefficients to be excluded at almost every weight. Boundary polynomials are included by ignoring the interior variables: the projection of  $\mathcal{H}_B$  onto  $\mathbb{R}^B$  is  $\mathbf{1}_B^\perp$ .

**Corollary 1.2.** *For generic positive weights, every nonconstant Steklov eigenfunction is injective on  $B$ . Its extension is injective on  $V_0$  and nonzero at every vertex of  $G$ . On each component of  $G - V_0$  the extension is constant, equal to its value at the attachment vertex.*

Indeed, apply Theorem 1.1 to the coordinate forms  $x_v$  and the differences  $x_v - x_z$ , for distinct  $v, z \in V_0$ , and then use Lemma 2.1. None of these forms vanishes identically on  $\mathcal{H}_B$ . The same theorem handles prescribed multiples, arithmetic progressions and higher-degree relations without separate perturbation constructions; some examples are given in Section 4.

## 1.2 Local variation of one eigenpair

The mechanism behind Theorem 1.1 is local. For  $u \in \mathbb{R}^V$ , an edge  $xy$  is *flat* if  $u(x) = u(y)$ . For  $F \subseteq E$ , define

$$E(u) = \{xy \in E : u(x) \neq u(y)\}, \quad c_F(u) = c(V, F \cap E(u)),$$

where  $c$  counts components, including isolated vertices. Write  $c(u) = c_E(u)$  and  $N = |V|$ . At a boundary vertex  $b_0$ , the normalization  $u(b_0) = 1$  gives the affine space

$$\mathcal{A}_{b_0}(G, B) = \left\{ v \in \mathbb{R}^V : \sum_{b \in B} v(b) = 0, \quad v(b_0) = 1 \right\}, \quad (1.2)$$

which has dimension  $N - 2$ .

**Theorem 1.3.** *Let  $w^0 > 0$ , let  $\lambda^0 > 0$  be a simple Steklov eigenvalue, and normalize its extension  $u^0$  by  $u^0(b_0) = 1$  at a boundary vertex where it is nonzero. Denote the corresponding local analytic eigenpair by  $(\lambda(w), u(w))$ .*

(i) *If only the weights in  $F \subseteq E$  vary, then*

$$\text{rank } D\Phi_F(w^0) = N - c_F(u^0), \quad \Phi_F : w|_F \mapsto (\lambda(w), u(w)). \quad (1.3)$$

*When all weights vary,  $\text{rank } D_w u(w^0) = N - c(u^0) - 1$ .*

(ii) *If a spanning tree  $T$  satisfies  $E(T) \subseteq E(u^0)$ , then, with all weights outside  $T$  fixed, the map*

$$w|_{E(T)} \mapsto (\lambda(w), u(w))$$

*is a local analytic diffeomorphism onto a neighbourhood of  $(\lambda^0, u^0)$  in  $(0, \infty) \times \mathcal{A}_{b_0}(G, B)$ . Its inverse is rational. The resulting weights remain positive, and the eigenvalue remains simple with the same ordered index. Fewer than  $N - 1$  variable edges cannot give a surjective differential onto this target.*

The rank formula applies at every simple positive eigenpair, including weights with flat edges and graphs with vertices outside the core. On the core, generic non-flatness will imply that every spanning tree of the core gives the local coordinates in part (ii). This is a statement about one eigenpair at a time. Eigenfunctions belonging to distinct eigenvalues are orthogonal on  $B$ , so several eigenpairs cannot be prescribed independently in this way.

Theorem 1.1 asserts that an extra algebraic equality is generically absent; Theorem 1.3 explains this by showing that the eigenpair image contains a relatively open set in its natural target. Neither assertion says that every point of that target is globally realizable by positive weights. In Section 5, we describe the remaining inequalities by directed cuts. This complementary result concerns the full extension and does not specify the index or simplicity of the realized eigenvalue.

### 1.3 Related work and proof outline

Poignard, Pereira and Pade [8] studied generic spectral properties of weighted graph Laplacians under perturbations that preserve the underlying graph. On metric graphs, generic simplicity, nonvanishing and further vertex conditions have been studied by Berkolaiko and Liu [2] and Alon [1]. In those results the variables are edge lengths, whereas here they are conductances. Our global realization problem is also distinct from the inverse Fiedler-vector problem considered by Lin and Shirazi [7], where the spectral position is part of the prescribed data.

We use the pendant-attachment formulas of Chen, Hua and Shi [3], and retain their attribution throughout. The additional point needed here is a tree construction for which every extension for a positive eigenvalue is non-flat on every tree edge. It uses cut sums rather than polynomial avoidance. After this construction, an edge-by-edge resultant argument gives generic non-flatness on the core. Section 3 proves the exact rank formula and an explicit inverse. Section 4 then proves Theorem 1.1 directly: local freedom first supplies a *single* weight avoiding all prescribed relations on all positive branches; a resultant makes the exceptional set algebraic. Joint relations involving the eigenvalue are subsequently obtained by homogenization and a common scaling of the weights. Thus the generic theorem is not used in the non-flatness construction on which its proof depends.

## 2 The boundary core and non-flat eigenfunctions

### 2.1 The Steklov matrix and the core

Order the vertices as  $V = I \sqcup B$  and write the blocks of  $L$  accordingly. The interior block  $L_{II}$  is positive definite. To see this, extend  $a \in \mathbb{R}^I$  by zero on  $B$ . Then

$$a^\top L_{II} a = \sum_{xy \in E} w(xy) (\tilde{a}(x) - \tilde{a}(y))^2.$$

If this expression is zero, connectedness makes  $\tilde{a}$  constant, hence zero. When  $I = \emptyset$ , the determinant of the empty block is understood to be 1.

Consequently, a boundary function  $f$  has the unique harmonic extension  $\mathcal{E}(w)f$  given by

$$(\mathcal{E}(w)f)|_B = f, \quad (\mathcal{E}(w)f)|_I = -L_{II}^{-1} L_{IB} f. \quad (2.1)$$

Its boundary flux is  $S(w)f$ , where

$$S(w) = L_{BB} - L_{BI} L_{II}^{-1} L_{IB}. \quad (2.2)$$

This is the Steklov matrix. The entries of  $S$  and  $\mathcal{E}$  are rational in  $w$ , with denominators powers of  $\Delta(w) = \det L_{II} > 0$ . Moreover,

$$f^\top S(w) f = (\mathcal{E}(w)f)^\top L(w) \mathcal{E}(w) f = \sum_{xy \in E} w(xy) ((\mathcal{E}f)(x) - (\mathcal{E}f)(y))^2.$$

Thus  $S$  is symmetric and positive semidefinite, with kernel  $\text{span}\{\mathbf{1}_B\}$ . Equation (1.1) is equivalent to  $S(w)f = \lambda f$  and  $u = \mathcal{E}(w)f$ . This proves the spectral assertions used in the introduction; see also [6].

**Lemma 2.1.** *The graph  $G^B$  is connected. Every component of  $G - V_0$  has exactly one neighbour in  $V_0$ . For every positive weight, harmonic extensions are constant on each such component. Their restriction to  $G^B$ , and the Steklov matrix, depend only on the weights in  $E(G^B)$ . Every edge of  $G$  with both endpoints in  $V_0$  belongs to  $G^B$ .*

*Proof.* The union of paths from one fixed boundary vertex to the other boundary vertices is connected. Every additional boundary-to-boundary path meets this union at its endpoints. Hence  $G^B$  is connected.

For any distinct  $x, y \in V_0$ , there are vertex-disjoint paths in  $G^B$  from  $x$  and  $y$  to distinct vertices of  $B$ , allowing paths of length zero. Otherwise the two-path form of Menger's theorem, applied in  $G^B$ , gives one vertex  $z$  meeting every path in  $G^B$  from  $\{x, y\}$  to  $B$ . Choose  $t \in \{x, y\} \setminus \{z\}$ . A simple boundary-to-boundary path through  $t$  lies in  $G^B$  and has two  $t$ - $B$  subpaths meeting only at  $t$ , so they cannot both contain  $z$ . This is a contradiction. In particular, an edge  $xy$  with both endpoints in  $V_0$ , together with these two paths, belongs to a simple boundary-to-boundary path.

Let  $K$  be a component of  $G - V_0$ . It has a neighbour in  $V_0$  by connectedness. If it had distinct neighbours  $x, y \in V_0$ , take a simple  $x$ - $y$  path through  $K$  and the two paths in  $G^B$  above. Those paths meet  $K$  only at  $x$  and  $y$ , so the concatenation is a simple boundary-to-boundary path meeting  $K$ . This contradicts the definition of  $V_0$ . Hence the attachment vertex is unique.

The Dirichlet problem on  $K$  has only one boundary value, at its attachment. The maximum principle makes its harmonic extension constant. Every edge outside the core therefore has equal endpoint values and contributes no current. Conversely, a harmonic function on the core extends constantly over all such components. Its boundary flux is unchanged, proving the remaining assertions.  $\square$

If  $G$  has a spanning tree whose leaves all lie in  $B$ , then  $G^B = G$ : every tree vertex lies on a boundary-to-boundary path, and the last assertion of Lemma 2.1 includes all remaining edges. The converse fails. For  $C_4$  with two opposite boundary vertices,  $G^B = G$ , but every spanning tree has an interior leaf. Thus replacing the core condition by the existence of such a spanning tree would unnecessarily restrict the results.

There is also no discrete unique-continuation principle that would make boundary zeros impossible for every weight. On the three-vertex path with  $B = V$  and unit weights, the vector  $(1, 0, -1)$  is an eigenvector of eigenvalue 1. The generic assertions must exclude exceptional weights.

## 2.2 Analytic branches and resultants

We record the perturbation fact used below, together with the short invertibility check that is relevant here; see also [10].

**Lemma 2.2.** *Let  $A(p)$  be a real symmetric matrix depending real analytically on finitely many parameters. Near a simple eigenpair  $(\lambda^0, f^0)$  at  $p^0$ , with  $f^0(b_0) = 1$ , there are unique local analytic branches  $(\lambda(p), f(p))$  satisfying  $f(p)(b_0) = 1$  and taking the prescribed values at  $p^0$ .*

*Proof.* Apply the analytic implicit-function theorem to

$$(p, t, f) \mapsto ((A(p) - tI)f, f(b_0) - 1).$$

The derivative in  $(t, f)$  sends  $(\eta, h)$  to  $((A(p^0) - \lambda^0 I)h - \eta f^0, h(b_0))$ . If this is zero, multiplication by  $(f^0)^\top$  gives  $\eta = 0$ . Simplicity then gives  $h = af^0$ , and  $h(b_0) = 0$  gives  $a = 0$ . This is an injective square linear map, hence invertible. The implicit-function theorem applies.  $\square$

Together with (2.1), this gives analytic normalized Steklov extensions. In particular, at a simple spectrum, any finite list of nonvanishing conditions on the branches persists under small changes of the weights.

For a symmetric matrix  $A$  and a simple eigenvalue  $t$  with eigenvector  $f$ ,

$$\text{adj}(A - tI) = c_t f f^\top, \quad c_t \neq 0. \tag{2.3}$$

For a repeated eigenvalue the adjugate is zero. Both assertions follow by diagonalizing  $A$ ; see [5].

Write  $p(t, w) = \det(S(w) - tI)$  and  $\tilde{p}(t, w) = p(t, w)/t$ . The roots of  $\tilde{p}$  at positive weights are exactly the positive Steklov eigenvalues, with multiplicity. Its leading coefficient is constant and nonzero. For a polynomial  $g(t, w)$  in  $t$  whose coefficients have denominators that are powers of  $\Delta(w)$ ,

$$\text{Res}_t(\tilde{p}, g) = \frac{R_g(w)}{\Delta(w)^{a_g}} \quad (2.4)$$

for a polynomial  $R_g$  and a nonnegative integer  $a_g$ . If a positive eigenvalue is a zero of  $g$ , then  $R_g(w) = 0$ . Conversely, at a positive weight for which  $g$  is nonzero at every positive eigenvalue, the resultant is nonzero. This follows directly from the product formula for the resultant over the roots of  $\tilde{p}$ .

The last observation is essential: writing a resultant does not by itself show that  $R_g$  is a nonzero polynomial. We must supply a weight at which *all* its factors are nonzero. Different relations may initially have different witnesses; a finite product of the resulting nonzero numerator polynomials then has a nonzero value somewhere in the positive orthant.

A nonzero polynomial has zero set of Lebesgue measure zero and empty interior. The measure assertion follows by induction on the number of variables and Fubini's theorem; the Hausdorff dimension is at most one less than the ambient dimension [4]. We also use the identity principle for real-analytic functions on a connected open set: a nonzero such function cannot vanish on an open subset. Its zero set is therefore closed with empty interior, and a finite union of these zero sets cannot fill an open ball.

### 2.3 A tree construction

On a path  $x_0 \cdots x_k$  with only its endpoints in the boundary, harmonicity says that the current  $w(x_{j-1}x_j)(u(x_{j-1}) - u(x_j))$  is independent of  $j$ . The path has effective weight

$$\alpha = \left( \sum_{j=1}^k \frac{1}{w(x_{j-1}x_j)} \right)^{-1}$$

and Steklov matrix  $\alpha \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ . Its extension is

$$u(x_j) = (1 - \theta_j)u(x_0) + \theta_j u(x_k), \quad \theta_j = \alpha \sum_{i=1}^j \frac{1}{w(x_{i-1}x_i)}. \quad (2.5)$$

For any fixed  $\alpha > 0$ , every  $0 = \theta_0 < \theta_1 < \cdots < \theta_k = 1$  is realized by

$$w(x_{j-1}x_j) = \frac{\alpha}{\theta_j - \theta_{j-1}}. \quad (2.6)$$

More generally, a path with no internal boundary vertex or external edges at its internal vertices can be replaced by an edge of weight  $\alpha$  without changing the Steklov matrix or the extension off the path. These are the path reductions of [3, Proposition 2.9 and Corollary 2.11].

We also use their pendant-attachment identity [3, Lemma 2.12]. We include a derivation in terms of the boundary flux.

**Lemma 2.3.** *Attach a new boundary leaf  $b$  to a vertex  $a$  by an edge of weight  $\tau > 0$ . If  $S$  is the old Steklov matrix, the new one is*

$$\hat{S} = \begin{pmatrix} S & 0 \\ 0 & 0 \end{pmatrix} + \kappa \begin{pmatrix} pp^\top & -p \\ -p^\top & 1 \end{pmatrix}. \quad (2.7)$$

Here  $p = e_a$  and  $\kappa = \tau$  for  $a \in B$ . For  $a \in I$ ,

$$p = -L_{BI}L_{II}^{-1}e_a, \quad \alpha = e_a^\top L_{II}^{-1}e_a, \quad \kappa = \frac{\tau}{1 + \tau\alpha}.$$

In both cases  $\mathbf{1}_B^\top p = 1$ .

*Proof.* For  $a \in B$ , the added flux is  $\tau(y(a) - z)$  at  $a$  and its negative at  $b$ , giving (2.7). Suppose  $a \in I$ , and prescribe old boundary values  $y$  and new value  $z$ . Write  $h = \mathcal{E}(w)y$  for the old extension and  $q = \tau(\hat{h}(a) - z)$  for the new current from  $a$  to  $b$ . On the old interior,

$$\hat{h}|_I = h|_I - qL_{II}^{-1}e_a.$$

Thus  $\hat{h}(a) = p^\top y - \alpha q$  and  $q = \kappa(p^\top y - z)$ . The old boundary flux becomes  $Sy + qp$ , while the flux at  $b$  is  $-q$ . This proves the matrix formula. Finally,  $p^\top \mathbf{1}_B = 1$  because the harmonic extension of constant boundary data is constant.  $\square$

If the old spectrum is simple and  $n = |B|$ , restrict (2.7) to  $\mathbf{1}_{n+1}^\perp$ . At  $\tau = 0$  the restriction has simple eigenvalues: the old positive ones and a simple zero with normalized eigenvector  $c(-\mathbf{1}_n, n)$ , where  $c = 1/\sqrt{n(n+1)}$ . Lemma 2.2 gives simple branches for all sufficiently small  $\tau > 0$ . The new smallest positive eigenvector tends to  $c(-\mathbf{1}_n, n)$ , and every other positive eigenvector tends to  $(f, 0)$  for an old eigenvector  $f$ . The old interior block remains invertible at  $\tau = 0$ , so the extensions on the old vertices converge as well.

Two further elementary observations will be useful. If a new positive boundary eigenvector  $(y, z)$  has  $z = 0$ , its last equation in (2.7) gives  $p^\top y = 0$ , and its first equation gives  $Sy = \lambda y$ . Thus such a zero is impossible whenever every old positive extension is nonzero at  $a$ . Also, on a tree  $T$ , summing (1.1) on one side of an edge  $ab$  gives

$$w(ab)(u(a) - u(b)) = \lambda \sum_{x \in B \cap V(T_a)} u(x), \quad (2.8)$$

where  $T_a$  is the component of  $T - ab$  containing  $a$ . All internal edge contributions cancel.

**Lemma 2.4.** *Let  $T$  be a tree whose leaves all lie in  $B$ , with  $|B| \geq 2$ . There are positive weights for which the Steklov spectrum is simple and every extension for a positive eigenvalue is nowhere zero and non-flat on every edge.*

*Proof.* Start with a path between two boundary vertices with no intermediate boundary vertex. If the current subtree does not contain  $B$ , choose a missing boundary vertex of minimum distance from the subtree and add its connecting path. The new path meets the old subtree only at one vertex and has no intermediate boundary vertex. The leaves of each intermediate subtree lie in its current boundary. The process ends with  $T$ , since a subtree containing every leaf of a tree is the whole tree.

For the initial path, the only positive eigenfunction has opposite nonzero boundary values. Formula (2.5) makes its extension strictly monotone. Choose the ratios in (2.6) so that no internal vertex has value zero.

Suppose suitable weights have been chosen on the old subtree with  $n$  boundary vertices. First contract the new path to a pendant edge of weight  $\tau$ . For sufficiently small  $\tau > 0$ , the new spectrum is simple. Each branch converging to an old positive eigenvalue is nowhere zero on the old vertices and non-flat on the old edges. Its value at the new leaf tends to zero, whereas its value at the attachment tends to a nonzero number. Hence the new edge is non-flat. The observation following Lemma 2.3 also excludes a zero at the new leaf.

For the new smallest positive branch, the extension tends to  $-c$  on every old vertex and to  $nc$  at the new leaf. Thus all contracted vertex values are nonzero for small  $\tau$ . Consider any edge and choose the component after its deletion that does not contain the new leaf. Each component of a tree minus one edge contains a leaf of the tree. In the chosen component that leaf is not the new leaf, hence it is an old boundary vertex. If there are  $m$  old boundary vertices in that component, its boundary sum tends to  $-cm \neq 0$ . Formula (2.8) therefore excludes a flat edge. One sufficiently small positive  $\tau$  works for every edge and every branch, since there are only finitely many of them.

Restore the new path using (2.6). Its endpoint values are distinct on every positive branch. For each internal vertex and each branch, a zero excludes at most one interpolation parameter. A finite union of these coordinate conditions cannot fill the open simplex of interpolation parameters. Choose parameters outside them. The Steklov matrix and old values are unchanged, all new edges are non-flat, and all new values are nonzero. This completes the induction.  $\square$

**Lemma 2.5.** *There is a nonzero polynomial  $P_*$  in the edge weights such that  $P_*(w) \neq 0$  implies a simple Steklov spectrum, nonzero boundary coordinates for every eigenfunction, and no flat core edge for any extension belonging to a positive eigenvalue.*

*Proof.* Fix a core edge  $e = xy$  and a simple boundary-to-boundary path containing it. Enlarge this path to a subtree  $T_*$  containing  $B$  by attaching paths to the missing boundary vertices. Every leaf of  $T_*$  lies in  $B$ . Lemma 2.4 gives weights on  $T_*$  with simple spectrum and  $u(x) \neq u(y)$  on every positive branch.

Extend  $T_*$  to a spanning tree  $T_0$  of  $G$ . Keep the chosen weights on  $T_*$ , give the remaining edges of  $T_0$  weight 1, and the edges outside  $T_0$  weight  $\varepsilon$ . At  $\varepsilon = 0$ , the edges of positive weight are exactly those of  $T_0$ . Every branch of  $T_0$  outside  $T_*$  contains no boundary vertex, so the extension is constant there and contributes no current. The Steklov matrix and the extension on  $T_*$  are therefore unchanged. The interior block is positive definite even at  $\varepsilon = 0$ . For sufficiently small  $\varepsilon > 0$ , simplicity, boundary nonvanishing and  $u(x) \neq u(y)$  persist on every positive branch. This is a positive witness for the fixed edge  $e$ .

Let  $a_e(w) = \mathcal{E}(w)^\top (e_x - e_y)$  and

$$g_e(t, w) = a_e(w)^\top \operatorname{adj}(S(w) - tI) a_e(w).$$

By (2.3), at a simple positive eigenvalue  $g_e$  vanishes exactly when the corresponding extension is flat on  $e$ . The witness above makes  $g_e$  nonzero at every positive eigenvalue. Hence the numerator  $R_e$  in (2.4) is not identically zero.

The same witness has simple spectrum and nonzero boundary coordinates on every branch. The numerator  $P_s$  of  $\operatorname{Res}_t(p, \partial_t p)$  is therefore nonzero, and its nonvanishing excludes repeated eigenvalues. For each  $b \in B$ , apply (2.4) to  $e_b^\top \operatorname{adj}(S - tI) e_b$ , obtaining a nonzero numerator  $R_b$ . Now

$$P_* = P_s \prod_{e \in E(G^B)} R_e \prod_{b \in B} R_b$$

is a nonzero polynomial. At a positive weight where this product does not vanish, all the required properties hold. Every eigenfunction for the zero eigenvalue is a nonzero constant and has no zero boundary coordinate.  $\square$

This proof uses neither Theorem 1.1 nor injectivity. It strengthens the pendant construction of [3] by controlling edge differences. Edges outside the core are flat for every weight by Lemma 2.1, so the lemma identifies precisely which edges can be non-flat generically.

### 3 Local variation and spanning-tree coordinates

Choose an orientation of  $E$  and let  $C$  be the incidence matrix, with column  $e_x - e_y$  for an edge directed from  $x$  to  $y$ . Then

$$L(w) = C \operatorname{diag}(w) C^\top. \quad (3.1)$$

For  $F \subseteq E$ , write  $C_F$  for the columns indexed by  $F$ . The orientation is only a notation: reversing an edge changes both its incidence column and its value difference by a sign.

*Proof of Theorem 1.3.* Write  $(\lambda, u) = (\lambda^0, u^0)$  and  $L = L(w^0)$ . The analytic branch follows from Lemma 2.2 and the harmonic extension formula. Its allowed value variations belong to

$$\mathcal{T} = \left\{ h \in \mathbb{R}^V : \sum_{b \in B} h(b) = 0, \quad h(b_0) = 0 \right\}.$$

Differentiating (1.1) gives

$$C \operatorname{diag}(C^\top u) \dot{w} = (\lambda M - L) \dot{u} + \dot{\lambda} M u. \quad (3.2)$$

The left side depends on the variable edges. To read its rank as the rank of the eigenpair map, we must first check that the right side loses no information about  $(\dot{\lambda}, \dot{u})$ .

Consider the linear map

$$\mathcal{K} : \mathbb{R} \times \mathcal{T} \longrightarrow \mathbf{1}_V^\perp, \quad \mathcal{K}(\eta, h) = (\lambda M - L)h + \eta M u.$$

Its image is in  $\mathbf{1}_V^\perp$ , since both  $u$  and  $h$  have zero boundary sum. If  $\mathcal{K}(\eta, h) = 0$ , multiplication by  $u^\top$  and symmetry give  $\eta u^\top M u = 0$ . The boundary trace of  $u$  is nonzero, so  $\eta = 0$ . Hence  $(L - \lambda M)h = 0$ .

Eliminating the interior variables identifies  $\ker(L - \lambda M)$  with  $\ker(S - \lambda I)$  by harmonic extension. Simplicity of  $\lambda$  gives  $h = au$ . The normalization gives  $h(b_0) = 0$  and  $u(b_0) = 1$ , so  $a = 0$ . Thus  $\mathcal{K}$  is injective. Both its domain and target have dimension  $N - 1$ , and it is therefore an isomorphism.

Restricting (3.2) to  $F$ , we obtain

$$\operatorname{rank} D\Phi_F(w^0) = \operatorname{rank} \left( C_F \operatorname{diag}((C^\top u)|_F) \right).$$

A flat edge gives a zero column, while every non-flat edge gives a nonzero multiple of its incidence column. A vector orthogonal to all the remaining columns is precisely a vector constant on each component of  $(V, F \cap E(u))$ . This orthogonal space has dimension  $c_F(u)$ , proving (1.3).

When all weights vary, scaling them gives  $\lambda(tw^0) = t\lambda$  and  $u(tw^0) = u$  for  $t$  near 1. The differential image therefore contains  $(\lambda, 0) \neq 0$ . The projection from  $\mathbb{R} \times \mathcal{T}$  onto  $\mathcal{T}$  has a one-dimensional kernel, contained in that image. It reduces the rank by exactly one, proving  $\operatorname{rank} D_w u = N - c(u) - 1$ .

For part (ii), let  $T$  be a non-flat spanning tree and write  $R = E \setminus E(T)$ . The map  $C_T : \mathbb{R}^{E(T)} \rightarrow \mathbf{1}_V^\perp$  is an isomorphism; denote its inverse by  $J_T$ . Keep the off-tree weights fixed and write  $L_R = C_R \operatorname{diag}(w^0|_R) C_R^\top$ . For a nearby target  $(\mu, v) \in (0, \infty) \times \mathcal{A}_{b_0}(G, B)$ , define

$$w_T(\mu, v) = \operatorname{diag}(C_T^\top v)^{-1} J_T (\mu M v - L_R v). \quad (3.3)$$

The argument of  $J_T$  has total sum zero. At  $(\lambda, u)$  this expression equals  $w^0|_{E(T)}$ . Its denominators are the differences along the tree edges, all nonzero at  $u$ . Thus the expression is rational and well-defined near  $(\lambda, u)$ , and its entries remain positive after the neighbourhood is reduced.

Substitution gives

$$L(w_T(\mu, v), w^0|_R)v = L_Rv + C_T J_T(\mu Mv - L_Rv) = \mu Mv.$$

The boundary trace is nonzero because  $v(b_0) = 1$ . Choose a small interval containing  $\lambda$  and no other Steklov eigenvalue at  $w^0$ . For all sufficiently nearby weights, this interval contains exactly one eigenvalue, counted with multiplicity. The value  $\mu$  lies in it after the target neighbourhood is reduced. Hence  $\mu$  remains simple and is the continuation of the original ordered branch. Conversely, the eigenpair equation determines the tree weights uniquely through (3.3). This proves the local analytic diffeomorphism. Its target has dimension  $1 + (N - 2) = N - 1$ , so fewer than  $N - 1$  variable edges cannot give a surjective differential.  $\square$

Formula (3.3) also applies when the off-tree weights vary. Consequently, whenever  $T$  is non-flat, the augmented map

$$w \longmapsto (\lambda(w), u(w), w|_R) \tag{3.4}$$

is a local analytic diffeomorphism. The extra coordinates are exactly the weights not needed to vary the eigenpair. We will use this observation for the probabilistic corollary in Section 4.

The local inverse permits  $\mu$  to remain fixed while  $v$  varies, or permits both to vary. Positivity is preserved because the target stays near an existing positive realization. No assertion about arbitrary distant targets is needed.

**Example 3.1.** Let  $G = 0 - 1 - 2 - 3 - 0$ , let  $B = \{0, 2\}$ , and assign the cyclic weights  $(1, 2, 3, 4)$ . A normalized positive eigenpair is

$$\lambda = \frac{100}{21}, \quad u = \left(1, -\frac{1}{3}, -1, \frac{1}{7}\right).$$

Keep  $w(30) = 4$  fixed and change the three other weights to

$$(w(01), w(12), w(23)) = \left(\frac{18}{13}, \frac{18}{7}, \frac{8}{3}\right).$$

Then  $\mu = 5$  and  $v = (1, -3/10, -1, 1/5)$  satisfy  $L(w)v = \mu Mv$ . The three variable edges form a spanning tree. Thus both interior values and the eigenvalue have changed while one edge weight stayed fixed. The boundary has two vertices, so there is only one positive eigenvalue.

## 4 Polynomial relations and their consequences

### 4.1 Homogeneous relations on the core

We now prove Theorem 1.1 without using the joint eigenvalue statement below. The local rank formula provides a common nonvanishing weight. The resultant is used only after this weight has been found.

*Proof of Theorem 1.1.* By Lemma 2.1, we may work on  $G^B$ . A polynomial in the core weights is also a polynomial in all weights, independent of the others. Thus assume  $G = G^B$ , write  $N = |V|$ , fix  $b_0 \in B$ , and abbreviate  $\mathcal{A} = \mathcal{A}_{b_0}(G, B)$ .

The positive orthant contains a weight at which the nonzero polynomial  $P_*$  of Lemma 2.5 does not vanish. Choose a small open ball  $U$  about that weight on which the spectrum is simple, every boundary coordinate is nonzero, and every extension for a positive eigenvalue is non-flat on all edges. Shrink  $U$  so that all normalized branches  $(\lambda_j, \hat{u}_j)$ , with  $\hat{u}_j(b_0) = 1$ , are analytic there.

A homogeneous  $Q$  that is nonzero on  $\mathcal{H}_B$  is nonzero on  $\mathcal{A}$ . Indeed, if  $Q$  vanished on  $\mathcal{A}$ , homogeneity would make it vanish at every point of  $\mathcal{H}_B$  with nonzero  $b_0$  coordinate. These points are dense in  $\mathcal{H}_B$ , which would make  $Q$  identically zero there.

Fix a spanning tree. At every point of  $U$  it is non-flat for every positive branch. Theorem 1.3(ii) shows that each map

$$\Phi_j : U \longrightarrow (0, \infty) \times \mathcal{A}, \quad w \longmapsto (\lambda_j(w), \hat{u}_j(w))$$

maps a neighbourhood of each point onto a set containing a relatively open neighbourhood in the target. Its projection  $w \mapsto \hat{u}_j(w)$  has the same local openness property into  $\mathcal{A}$ . Hence  $Q \circ \hat{u}_j$  cannot vanish on a nonempty open subset of  $U$ : otherwise  $Q$  would vanish on an open subset of  $\mathcal{A}$ , and therefore identically on that affine space.

There are finitely many pairs  $(Q, j)$ . Their analytic zero sets are closed with empty interior in  $U$ , so they cannot cover  $U$ . Choose *one* weight  $w^* \in U$  at which  $Q(\hat{u}_j(w^*)) \neq 0$  for every  $Q \in \mathcal{F}$  and every positive branch  $j$ . This simultaneous choice is what allows the resultant to test all positive eigenvalues at once.

To construct the exceptional polynomial, write

$$z(t, w) = \mathcal{E}(w) \operatorname{adj}(S(w) - tI)e_{b_0}. \quad (4.1)$$

At a simple positive eigenvalue with nonzero  $b_0$  coordinate, (2.3) gives

$$z_{b_0}(\lambda_j, w) \neq 0, \quad \frac{z(\lambda_j, w)}{z_{b_0}(\lambda_j, w)} = \hat{u}_j(w). \quad (4.2)$$

For a homogeneous  $Q$  of degree  $d$ , the function  $g_Q(t, w) = Q(z(t, w))$  is polynomial in  $t$  with rational coefficients in  $w$ . At the common weight  $w^*$ ,

$$g_Q(\lambda_j, w^*) = z_{b_0}(\lambda_j, w^*)^d Q(\hat{u}_j(w^*)) \neq 0$$

for every  $j$ . Therefore

$$\operatorname{Res}_t(\tilde{p}(t, w), g_Q(t, w)) = \frac{R_Q(w)}{\Delta(w)^{a_Q}}$$

has a numerator  $R_Q \neq 0$ . The polynomial  $P_* \prod_{Q \in \mathcal{F}} R_Q$  has the required property. Outside its zero set, any nonzero scaling  $u$  of a normalized positive branch satisfies

$$Q(u) = u(b_0)^d Q(\hat{u}) \neq 0.$$

Nonzero constant polynomials cause no difficulty, and for the empty family we use  $P_*$  alone.

For an at most countable family, apply the finite assertion to each member separately. The exceptional set is contained in a countable union of polynomial zero sets, hence is null. Finally, a polynomial vanishes on  $\mathcal{H}_B$  if and only if it is divisible by  $s(x) = \sum_{b \in B} x_b$ : division by this linear polynomial leaves a remainder in the other coordinates, which is exactly the restriction to the hyperplane after one boundary variable is eliminated. Multiples of  $s$  vanish on every positive extension. Every other homogeneous polynomial is excluded at some positive weight by the finite assertion, and so is not a universal identity.  $\square$

**Corollary 4.1.** *Let  $\mathcal{F}$  be a finite family of homogeneous polynomials in the boundary variables, none vanishing identically on  $\mathbf{1}_B^\perp$ . For generic positive weights, every eigenfunction belonging to a positive eigenvalue avoids every polynomial in  $\mathcal{F}$ . The same assertion holds outside a countable union of polynomial zero sets for an at most countable family.*

*The only universal linear identities on the core have coefficient vector in  $\mathbb{R}\mathbf{1}_B$ , where  $\mathbf{1}_B$  is extended by zero off  $B$ .*

*Proof.* The projection of  $\mathcal{H}_B$  onto the boundary coordinates is all of  $\mathbf{1}_B^\perp$ . Thus a boundary polynomial satisfying the hypothesis remains nonzero when regarded as a polynomial on  $\mathcal{H}_B$ , and Theorem 1.1 applies. A linear form vanishes on  $\mathcal{H}_B$  exactly when its coefficient vector belongs to  $\mathcal{H}_B^\perp = \mathbb{R}\mathbf{1}_B$ .  $\square$

The injectivity assertion in Corollary 1.2 is now proved as well. More generally, for any finite  $K \subset \mathbb{R}$ , generic weights give

$$\sum_{v \in V_0} a_v u(v) \neq 0 \quad (a \in K^{V_0} \setminus \mathbb{R}\mathbf{1}_B)$$

for every extension belonging to a positive eigenvalue. Thus the only exception for a nonzero relation is full boundary support with equal coefficients there and zero coefficients at the interior vertices.

For example, on the boundary, any prescribed finite set of multipliers  $k$  gives  $kf(i) \neq f(j)$  for  $i \neq j$ , except for the forced relation with  $|B| = 2$  and  $k = -1$ . If  $|B| \geq 3$ , opposite values are excluded, so the boundary values of  $|f|$  are pairwise distinct. If  $|B| \geq 4$ , then  $f(i) + f(k) \neq 2f(j)$  for pairwise distinct  $i, j, k$ . Each assertion uses only a finite family of linear forms.

Higher-degree examples need no new perturbation argument. The following table lists polynomials that are nonzero on the boundary zero-sum hyperplane under the indicated sufficient hypotheses. All displayed vertices in a row are pairwise distinct.

| Polynomial               | Sufficient boundary size |
|--------------------------|--------------------------|
| $x_i^2 - x_j x_k$        | $ B  \geq 3$             |
| $x_i x_j - x_k x_\ell$   | $ B  \geq 5$             |
| $x_i^2 + x_j^2 - x_k^2$  | $ B  \geq 4$             |
| $x_i^3 - x_j^2 x_k$      | $ B  \geq 3$             |
| $x_i x_j x_k - x_\ell^3$ | $ B  \geq 4$             |

Whenever a further boundary coordinate is available, it absorbs the zero-sum constraint and leaves the displayed variables free. For the first and fourth rows with three boundary vertices, substituting  $x_k = -x_i - x_j$  gives respectively  $x_i^2 + x_i x_j + x_j^2$  and  $x_i^3 + x_i x_j^2 + x_j^3$ , neither identically zero. For the last row with four boundary vertices, the restriction is  $xyz + (x + y + z)^3$ , also nonzero.

The same examples hold for vertices of the core. If a boundary coordinate is not displayed, it absorbs the constraint. If all boundary coordinates are displayed, the stated lower bounds force the displayed vertices to be exactly the boundary, reducing to the preceding checks. This presentation is not meant to count each example as a separate phenomenon. Some relations reduce to already excluded ones in small dimensions; the theorem handles all of them through the same hyperplane condition.

## 4.2 Normalized relations involving the eigenvalue

Write  $\mathcal{A}_{b_0}^B = \mathcal{A}_{b_0}(G^B, B)$ . At a simple spectrum with nonzero boundary coordinates, let  $0 < \lambda_1 < \dots < \lambda_{|B|-1}$  and normalize the core extensions by  $\hat{u}_j(b_0) = 1$ . Normalization makes nonhomogeneous relations meaningful. The inclusion of  $\lambda_j$  in such relations requires no further rank calculation: a common scaling of the weights leaves the normalized extensions unchanged and scales all the eigenvalues.

**Corollary 4.2.** *Fix  $b_0 \in B$  and let  $\mathcal{F} \subset \mathbb{R}[t, x_v : v \in V_0]$  be finite. Suppose no member vanishes identically on  $(0, \infty) \times \mathcal{A}_{b_0}^B$ . There is a nonzero polynomial  $R_{\mathcal{F}}$  in the edge weights such that  $R_{\mathcal{F}}(w) \neq 0$  implies simple spectrum, nonzero boundary coordinates, and*

$$P(\lambda_j, \hat{u}_j) \neq 0 \quad (P \in \mathcal{F}, 1 \leq j \leq |B| - 1).$$

*For an at most countable family, the exceptional set is contained in a countable union of zero sets of nonzero polynomials.*

*Proof.* As before, work on  $G^B$ . For each  $P \in \mathcal{F}$ , write

$$P(t, x) = \sum_{r=0}^{m_P} t^r p_{P,r}(x).$$

At least one coefficient, say  $p_{P,r_P}$ , is not identically zero on  $\mathcal{A}_{b_0}^B$ . With  $d_P = \deg p_{P,r_P}$ , its homogenization is the polynomial

$$Q_P(x) = x_{b_0}^{d_P} p_{P,r_P}(x/x_{b_0}),$$

where the right side is expanded as a polynomial, so it is defined also at  $x_{b_0} = 0$ . On  $\mathcal{A}_{b_0}^B$  it equals  $p_{P,r_P}$ , and hence it does not vanish identically on  $\mathcal{H}_B$ .

Theorem 1.1, together with Lemma 2.5, gives a positive weight  $w^0$  at which every normalized positive branch satisfies  $p_{P,r_P}(\hat{u}_j) \neq 0$  for all  $P \in \mathcal{F}$ . At the scaled weights  $cw^0$ ,  $c > 0$ , the extension  $\hat{u}_j$  is unchanged and the eigenvalue is  $c\lambda_j$ . Consequently,

$$c \mapsto P(c\lambda_j, \hat{u}_j)$$

is a nonzero polynomial: its coefficient of  $c^{r_P}$  is  $\lambda_j^{r_P} p_{P,r_P}(\hat{u}_j) \neq 0$ . Each pair  $(P, j)$  excludes only finitely many positive  $c$ . Choose a common  $c$  outside their union. The weight  $w^* = cw^0$  avoids every prescribed relation on every positive branch.

This witness lies on one scaling ray. To obtain the claimed algebraic exceptional set in the whole weight space, use the resultant once more. For  $P(t, x) = \sum_{\alpha} p_{\alpha}(t)x^{\alpha}$  and  $d = \max_{p_{\alpha} \neq 0} |\alpha|$ , let

$$P^h(t, z) = \sum_{\alpha} p_{\alpha}(t) z^{\alpha} z_{b_0}^{d-|\alpha|}.$$

Use  $z(t, w)$  from (4.1). At the common witness, (4.2) gives

$$P^h(\lambda_j, z(\lambda_j, w^*)) = z_{b_0}(\lambda_j, w^*)^d P(\lambda_j, \hat{u}_j) \neq 0.$$

Thus  $\text{Res}_t(\tilde{p}, P^h(t, z(t, w)))$  has a nonzero numerator  $R_P(w)$ . The product  $P_* \prod_{P \in \mathcal{F}} R_P$  proves the finite assertion. For an at most countable family, apply this assertion one polynomial at a time.  $\square$

This proof distinguishes the two roles of scaling and elimination. Scaling supplies a common nonvanishing point for the new relations; it is not itself a full-dimensional genericity argument. The resultant supplies that final step.

**Corollary 4.3.** *The polynomials in  $\mathbb{R}[t, x_v : v \in V_0]$  that vanish for every positive weight and every positive eigenpair for which the normalization at  $b_0$  is defined form the ideal*

$$\left\langle \sum_{b \in B} x_b, x_{b_0} - 1 \right\rangle. \quad (4.3)$$

*For any  $b_1 \in B \setminus \{b_0\}$ , almost every positive weight has the following property: for each fixed spectral index  $j$ , the  $N_0 - 1$  numbers*

$$\lambda_j, \quad \hat{u}_j(v) \quad (v \in V_0 \setminus \{b_0, b_1\})$$

*are algebraically independent over  $\mathbb{Q}$ .*

*Proof.* Every normalized pair satisfies the two affine equations in (4.3). Conversely, substitute

$$x_{b_0} = 1, \quad x_{b_1} = -1 - \sum_{b \in B \setminus \{b_0, b_1\}} x_b.$$

The quotient by the displayed ideal is the polynomial ring in  $t$  and the remaining  $N_0 - 2$  vertex variables. A polynomial outside the ideal is therefore nonzero on  $(0, \infty) \times \mathcal{A}_{b_0}^B$  and, by Corollary 4.2, fails to be a universal identity. For the last assertion, apply the countable version of that corollary to all nonzero rational-coefficient polynomials in these free coordinates. The finitely many spectral indices are already covered simultaneously.  $\square$

The final assertion concerns each eigenpair separately. It does not assert algebraic independence among the coordinates of several different eigenpairs. Their boundary traces satisfy orthogonality relations.

### 4.3 Absolutely continuous random weights

The local coordinates have a measure-theoretic consequence that does not depend on independence of the edge weights.

**Corollary 4.4.** *Suppose the positive edge weights have a joint distribution absolutely continuous with respect to Lebesgue measure on  $\mathbb{R}_{>0}^E$ . For each  $1 \leq j \leq |B| - 1$ , the law of  $(\lambda_j, \hat{u}_j)$  is absolutely continuous with respect to the  $(N_0 - 1)$ -dimensional Lebesgue measure on  $(0, \infty) \times \mathcal{A}_{b_0}^B$ .*

*Proof.* The core eigenpair depends only on the core weights. Their marginal distribution is absolutely continuous by Fubini's theorem, so assume  $G = G^B$  and write  $m = |E|$ ,  $N = |V|$ . Outside the null set defined by  $P_*$ , the ordered eigenpairs are simple, their normalization is defined, and every spanning tree is non-flat for every positive branch.

Fix  $j$  and a spanning tree  $T$ , and write  $R = E \setminus E(T)$ . In a neighbourhood of each nonexceptional weight, (3.4) gives a local analytic diffeomorphism

$$w \mapsto (\lambda_j(w), \hat{u}_j(w), w|_R)$$

onto an open subset of  $(\mathbb{R} \times \mathcal{A}_{b_0}(G, B)) \times \mathbb{R}^{m-N+1}$ . Use fixed affine coordinates on  $\mathcal{A}_{b_0}(G, B)$ . These neighbourhoods have a countable subcover, since the weight space is second countable.

Let  $Z$  be a Lebesgue null set in  $(0, \infty) \times \mathcal{A}_{b_0}(G, B)$ . By Fubini's theorem,  $Z \times \mathbb{R}^{m-N+1}$  is null. The inverse of each coordinate diffeomorphism is  $C^1$  and is Lipschitz on sufficiently small compact neighbourhoods. A countable cover by such neighbourhoods shows that it maps null sets to null sets. Thus the preimage of  $Z$  is null in every chart, hence in their countable union. The omitted algebraic set is null as well. Absolute continuity of the weight law now gives probability zero to the full preimage of  $Z$ .  $\square$

The normalization and ordering are defined almost surely; arbitrary definitions on the exceptional null set do not affect the conclusion. No smoothness or boundedness of the resulting density is asserted.

## 5 Realization by positive weights

The local-coordinate theorem describes a neighbourhood of an existing simple eigenpair. It does not make the entire normalized affine space realizable. This distinction already appears on a path.

**Example 5.1.** Let  $G = s - x - t$  and  $B = \{s, t\}$ . Prescribe  $v(s) = 1$ ,  $v(t) = -1$ , and  $\lambda > 0$ . The boundary equations force

$$w(sx) = \frac{\lambda}{1 - v(x)}, \quad w(xt) = \frac{\lambda}{1 + v(x)}.$$

These weights are positive exactly when  $-1 < v(x) < 1$ , and in that interval they also make  $v$  harmonic at  $x$ . The feasible interval contains an open set and has no additional polynomial identity, but is not the whole line of possible interior values.

We give a general description of these positivity restrictions. The prescribed data now include the values at *every* vertex, not only the boundary trace. The function need not be harmonic before the weights are chosen, and the realized eigenvalue is not required to have a specified index or to be simple.

For  $v \in \mathbb{R}^V$ , direct each non-flat edge from its larger value to its smaller value and denote the resulting directed graph by  $D_v$ . Flat edges are omitted. Its underlying spanning graph  $(V, E(v))$  may be disconnected. For  $X \subseteq V$ , write

$$s_B(v; X) = \sum_{b \in B \cap X} v(b).$$

The relevant condition is a strict form of the positive-flow criterion. We include its proof, using the nonnegative version of Farkas' lemma [9].

**Lemma 5.2.** *Let  $D$  be a finite directed graph with connected underlying graph, and let  $C$  be its incidence matrix, positive at each tail. Let  $b \in \mathbb{R}^{V(D)}$  have total sum zero. There is a strictly positive edge vector  $q$  with  $Cq = b$  if and only if  $b(X) > 0$  for every nonempty proper vertex set  $X$  with no incoming edge. For a single vertex without edges, the condition means  $b = 0$  and the edge vector is empty.*

*Proof.* If  $Cq = b$  and  $X$  has no incoming edge, then

$$b(X) = \sum_{e \in \delta^+(X)} q_e > 0.$$

The last inequality follows because a nonempty proper set in a connected underlying graph has a cut edge, which here must be outgoing.

Conversely, the one-vertex case is immediate. In every other case, choose  $\varepsilon > 0$  small enough that

$$b(X) - \varepsilon |\delta^+(X)| \geq 0$$

for every nonempty proper set without incoming edges. There are finitely many such sets; if there are none, any  $\varepsilon > 0$  will do. The vector  $b' = b - \varepsilon C\mathbf{1}$  has total sum zero and  $b'(X) \geq 0$  for all sets without incoming edges, including the empty and full sets.

To apply Farkas' lemma, let  $z \in \mathbb{R}^{V(D)}$  satisfy  $C^\top z \geq 0$ . Along every edge, the value at the tail is at least the value at the head. Each upper level set of  $z$  therefore has no incoming edge. If  $a_1 < \dots < a_s$  are the distinct values of  $z$ , then

$$z = a_1 \mathbf{1} + \sum_{i=2}^s (a_i - a_{i-1}) \mathbf{1}_{X_i}, \quad X_i = \{x : z(x) \geq a_i\}.$$

It follows that  $z^\top b' \geq 0$ . Farkas' lemma now gives  $q' \geq 0$  with  $Cq' = b'$ . The vector  $q = q' + \varepsilon \mathbf{1}$  is strictly positive and satisfies  $Cq = b$ .  $\square$

**Proposition 5.3.** *Let  $v|_B \neq 0$ ,  $s_B(v; V) = 0$ , and  $\lambda > 0$ . There exist positive weights  $w$  satisfying  $L(w)v = \lambda Mv$  if and only if each component  $K$  of  $(V, E(v))$  satisfies*

$$(i) \quad s_B(v; K) = 0;$$

$$(ii) \quad s_B(v; X) > 0 \text{ whenever } \emptyset \neq X \subsetneq K \text{ and } D_v \text{ has no edge from } K \setminus X \text{ into } X.$$

*When these conditions hold, the weights on flat edges may be chosen arbitrarily. For any one positive realization  $w^0$ , all positive realizing weights form the relatively open convex polyhedral set*

$$\mathcal{W}_{\lambda, v} = (w^0 + \ker(C \operatorname{diag}(C^\top v))) \cap \mathbb{R}_{>0}^E, \quad (5.1)$$

*where  $C$  is an incidence matrix of  $G$ . This set has dimension  $|E| - |V| + c(v)$ .*

*Proof.* Let  $C_v$  be the incidence matrix of the non-flat edges oriented as in  $D_v$ . For a positive realization, the currents

$$q_{ab} = w(ab)(v(a) - v(b)) \quad (a \rightarrow b \text{ in } D_v)$$

are strictly positive. Flat edges contribute zero, so the eigenpair equation becomes

$$C_v q = \lambda Mv. \quad (5.2)$$

Summing over a component  $K$  gives (i). If  $X$  has no incoming edge in  $K$ , summing over  $X$  gives

$$\lambda s_B(v; X) = \sum_{e \in \delta^+(X)} q_e > 0.$$

For a nonempty proper  $X$ , connectedness within  $K$  supplies an outgoing cut edge. This proves (ii).

For sufficiency, apply Lemma 5.2 in each component, with source vector  $(\lambda Mv)|_K$ . Conditions (i) and (ii) are exactly its hypotheses. At an isolated vertex, (i) makes the source zero. We obtain positive currents on all non-flat edges satisfying (5.2). Define

$$w(ab) = \frac{q_{ab}}{v(a) - v(b)} \quad (a \rightarrow b \text{ in } D_v)$$

and give every flat edge any positive weight. The resulting weights satisfy  $L(w)v = \lambda Mv$ . Since  $v|_B \neq 0$ , this is a Steklov eigenpair.

For fixed  $(\lambda, v)$ , the equation is linear in the weights:

$$C \operatorname{diag}(C^\top v)w = \lambda Mv.$$

Once  $w^0$  is a solution, all positive solutions are precisely (5.1). The coefficient matrix has rank  $|V| - c(v)$ , by the same incidence-matrix argument as in Theorem 1.3. Its affine solution space has dimension  $|E| - |V| + c(v)$ . Intersecting with the positive orthant preserves this dimension, because the intersection contains a relatively open neighbourhood of  $w^0$  in that affine space. The flat-edge columns are zero, which also proves directly that those weights are unrestricted.  $\square$

**Corollary 5.4.** *The functions that are non-flat on every edge and realizable as extensions for positive Steklov eigenvalues form a finite union of relatively open convex polyhedral cones in  $\{v : \sum_{b \in B} v(b) = 0\}$ . For a fixed realizable pair  $(\lambda, v)$  of this kind, the realizing weights have dimension  $|E| - |V| + 1$ .*

*If  $G = T$  is a tree and  $v$  is non-flat on every edge, a prescribed pair with  $\lambda > 0$ ,  $v|_B \neq 0$  and zero boundary sum has at most one positive realization. Its only possible weights are*

$$w(ab) = \lambda \frac{s_B(v; V(T_a))}{v(a) - v(b)}, \quad (5.3)$$

*where  $T_a$  is the component of  $T - ab$  containing  $a$ . These weights realize the pair if and only if they are all positive.*

*Proof.* For each orientation  $D$  of  $E$ , impose the zero boundary sum, the inequalities  $v(a) > v(b)$  on all directed edges  $a \rightarrow b$ , and the inequalities  $s_B(v; X) > 0$  on nonempty proper sets without incoming edges in  $D$ . These homogeneous linear equalities and strict inequalities define a relatively open convex polyhedral cone, possibly empty. By Proposition 5.3, their union over the finitely many orientations is precisely the desired set. A nonempty such cone has nonzero boundary trace. Strict decrease along the edges makes  $D$  acyclic, so some vertex  $s$  has no incoming edge. The inequality  $s_B(v; \{s\}) > 0$  then forces  $s \in B$  and  $v(s) > 0$ . For a non-flat function,  $c(v) = 1$ , giving the stated dimension.

On a tree, summing the eigenpair equation over  $T_a$  gives (5.3), so the weights are uniquely determined. Conversely, orient each edge  $ab$  from  $a$  to  $b$ . The incidence map  $C : \mathbb{R}^{E(T)} \rightarrow \mathbf{1}_V^\perp$  is an isomorphism. Its unique solution of  $Cj = \lambda Mv$  has  $j_{ab} = \lambda s_B(v; V(T_a))$ , as follows by summing the divergence over  $T_a$ . Therefore, if all weights in (5.3) are positive, they satisfy  $C \operatorname{diag}(C^\top v)w = \lambda Mv$  and realize the prescribed pair.  $\square$

For a function non-flat on every edge, the fibre also has the form

$$\mathcal{W}_{\lambda, v} = \left( w^0 + \operatorname{diag}(C^\top v)^{-1} \ker C \right) \cap \mathbb{R}_{>0}^E.$$

Thus the freedom in the weights is a rescaled circulation space; tree uniqueness is the case in which this space is zero. These are full-extension statements, not recovery from a boundary eigenvector alone.

Finally, the cut criterion does not depend on the chosen positive value of  $\lambda$ , since multiplying all weights by a positive constant multiplies every Steklov eigenvalue by that constant. It does not distinguish spectral indices or assert simplicity. Those properties are retained in the local inverse of Theorem 1.3 because the target stays near an existing simple branch. This explains both the relation and the difference between the local-coordinate theorem and the global realization criterion.

**Declaration on the use of generative AI.** Generative AI tools were used in drafting and revising the exposition and in developing parts of the arguments.

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